

Drought and postdrought recovery cycles in an intermittent Mediterranean stream: structural and functional aspects

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Abstract. The effects of the intensity of seasonal droughts on stream ecosystems were studied in the Fuirosos, an intermittent forested Mediterranean stream. Macroinvertebrate community structure and stream ecosystem metabolism were measured during seasonal summer droughts in 2001, 2002, and 2003. Ecosystem metabolism was profoundly affected by stream intermittency. Organic matter that accumulated during the dry period enhanced ecosystem respiration during the postdrought recovery. Highest biotic diversity was found at low water levels as the stream dried and contracted. Macroinvertebrate community response to drying was stepped and apparently defined by thresholds of transition from drying to cessation of flow and from the dry phase to restoration of flow. Environmental conditions changed markedly with cessation of flow, causing large changes in community structure during 2001 and 2003 (dry years). Drying caused an increase in macroinvertebrate density that peaked in isolated pools soon after flow ceased, but then decreased rapidly because of the physicochemical changes associated with fragmentation of the watercourse. The macroinvertebrate community at the end of the summer dry phase (when flow resumed) differed from the community that had been present before drying began. Differences in community structure during the summer dry period were not as marked in 2002 (a wet year) as in 2001/2003. The influence of drought on the macroinvertebrate community differed across substrata. Drying led to significant changes in density on cobbles and leaves, but not on sand. Few taxa resisted drying, and resilience to drying was the dominant response to disturbance in the Fuirosos.

Key words: ecosystem contraction, drought effect, postdrought recovery, macroinvertebrate community structure, stream metabolism, Mediterranean streams.

Stream ecosystems in Mediterranean climates are hydrologically diverse and highly dynamic. Flows range from high-discharge floods to periods of complete drought. Much is known about how floods shape ecological processes, but far less is known about the biological and chemical changes that occur during periods of drought (Stanley et al. 1997). Droughts are characterized by a sequential decline in precipitation, runoff, soil moisture, groundwater levels, and stream flow that may involve the cessation

of surface flow (Dahm et al. 2003). Droughts in Mediterranean regions are predictable and periodic, and they are typified as seasonal droughts (Lake 2003). However, drought intensity can vary from year to year because of interannual variability in weather. Wet and dry years alternate, and the stream channel may dry up partially or completely depending on the year (Sabater et al. 2001, Butturini et al. 2003). Studies of the effects of drought on Mediterranean stream ecosystems have focused mainly on invertebrate responses to drying (e.g., Towns 1985, Gasith and Resh 1999, Pires et al. 2000, Bravo et al. 2001), and little is known about effects on ecosystem processes (Grimm and Petrone 1997, Maamri et al. 1997, Román and Sabater 1997, Dent et al. 2001, Acuña et al. 2004), especially ecosystem metabolism.

The physical effect of drying has been described as ecosystem contraction, whereas flow

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recovery has been described as ecosystem expansion (Stanley et al. 1997, Ward and Tockner 2001). In this sense, intermittent Mediterranean streams may be characterized as spatially dynamic ecosystems that experience cycles of expansion, contraction, and fragmentation. These cycles induce rapid changes in ecosystem size that are a fundamental property of lotic systems (de Vries 1995, Stanley et al. 1997), but this property often is neglected by stream ecologists.

Ecological changes during the drying process may be gradual at first, but cessation of flow causes abrupt loss of habitat, alteration of physicochemical conditions in pools, and fragmentation of the watercourse. Thus, many ecological responses to drought are stepped—gradual changes alternate with thresholds of swift transitions (e.g., a habitat disappears or is fragmented) (Boulton 2003, Lake 2003). However, it is not clear which responses are gradual ramps and which responses occur as irregular steps (Lake 2003), and further research is necessary to clarify these patterns (Boulton 2003).

We assessed the effects of seasonal drought on ecosystem structure and function in a Mediterranean stream. We hypothesized that drought intensity would affect organic matter dynamics of the stream ecosystem by altering stream transport capacity during leaf fall. We predicted that drought would cause discontinuities or thresholds rather than gradual changes in the macroinvertebrate community structure, and that ecosystem contraction would induce maximum habitat heterogeneity and biotic diversity at some intermediate water level.

Methods

Study area

The study site was in a 16-km² granitic catchment (Fuirosos) at an altitude of 150 m above sea level near Barcelona in northeastern Spain (lat 41°42'N, long 2°34'E). Nearly the whole catchment is included in the Montnegre–Corredor Natural Park, and holm oak, coniferous, and deciduous forest covers 90% of the total catchment (Fig. 1). The climate is Mediterranean with mild winters and warm springs and summers. Monthly average air temperatures range from 4°C in December to 28°C in July and August. Air temperatures <0°C are infrequent during winter. Highest precipitation occurs most often

in autumn and spring (Fig. 2), with occasional storms in summer. The high variability in rainfall that is typical of Mediterranean systems is evident in deviations from mean monthly precipitation and high interannual differences in rainfall (Table 1).

The Fuirosos is an intermittent 3rd-order stream. Water flow usually ceases from July/August to September/October (Sabater et al. 2001). At a discharge of 30 L/s, the stream was 3 to 4 m wide with depths ranging from 0.1 to 0.2 m in riffles and up to 0.4 m in pools. A 50-m downstream reach was selected for study in a riffle section of uniform channel morphology and channel slope (0.9%). The reach has a well-developed riparian zone that is 10 to 20 m wide and forms a closed canopy from May to October. The riparian vegetation is dominated by alder (*Alnus glutinosa* L.), hazelnut (*Corylus avellana* L.), and plane (*Platanus acerifolia* Aiton).

Physicochemical characteristics

Water level was monitored continuously during the study in one of the more permanent pools using a pressure transducer (PDCR 1830, Druck Limited, Groby, Leicester, UK), which was connected to an automatic sampler (Sigma 900 Max, Hach Company, Loveland, Colorado). Discharge and average velocity were determined biweekly from 1 June to 1 November 2001, 2002, and 2003 using the slug-injection method with Cl⁻ as a conservative tracer (Gordon et al. 1992). These measurements and the corresponding stage provided an empirical stage–discharge relationship that was used to calculate discharge based on the water-level record (Bernal et al. 2002). Depth, width, and length of the remaining pools were measured every 3 to 4 d after the onset of stream fragmentation.

Conductivity and pH were measured in situ (MultiLine P4, Wissenschaftlich-Technische Werkstätten, Weilheim, Germany). Stream water for nutrient analyses was filtered in situ through glass-fiber filters (Whatman GF/F) and stored at 4°C until analysis. NO₃-N and NH₄-N were analyzed using a Skalar autoanalyzer (SKALAR, Breda, The Netherlands), and PO₄-P was analyzed with a Perkin–Elmer spectrophotometer (APHA 1992).

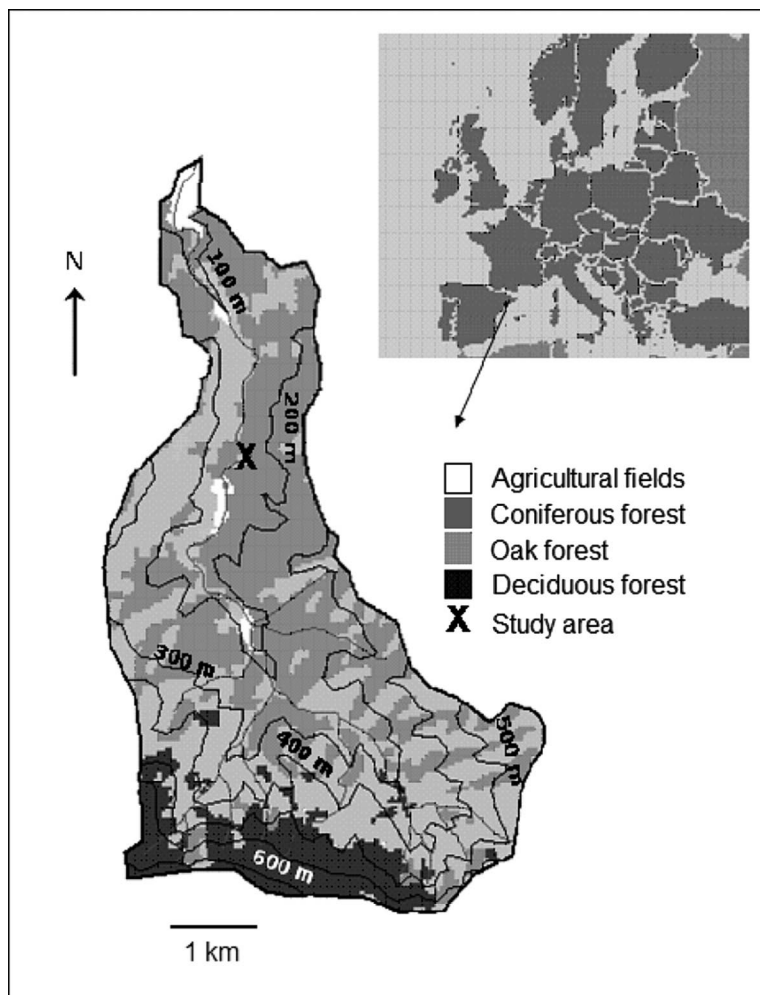


FIG. 1. The Fuirosos catchment.

Streambed mapping

The distribution of substrata in the streambed was mapped monthly along 11 equidistant transects. Percentage of the streambed covered by different inorganic substrates (rock, cobble, sand, and gravel) and different types of benthic organic matter (BOM; leaves, branches, or fine detritus) were estimated. Percentages of each substratum along all transects were averaged and used to calculate BOM cover and macroinvertebrate density and biomass at the reach scale.

BOM

Triplicate samples of BOM were collected monthly at random points over the streambed

with a 20-cm-diameter corer. Samples were dried at 105°C to constant mass. C and N content were measured by organic elemental analysis using combustion and gas chromatography (EA 1108 CHNS-O, Carlo-Erba Instruments, Milan, Italy).

Stream ecosystem metabolism

Whole-ecosystem gross primary production (GPP) and ecosystem respiration (ER) were determined by the single-station diel O₂ method (Odum 1956) (see detailed methods in Acuña et al. 2004). Dissolved O₂ (DO) concentration was recorded continuously during 24 periods of 36 h between June and November 2001 and 2002

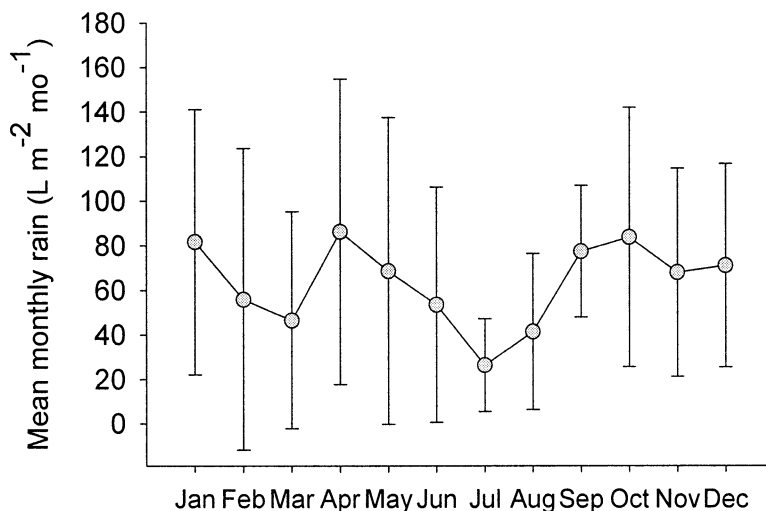


FIG. 2. Mean (± 1 SD) monthly accumulated rain in the Fuirosos from October 1996 to April 2004 as recorded at the Hortsavinyà meteorological station at the southern limit of the catchment.

(Acuña et al. 2004). DO (Sensorex CS 511, Campbell Scientific, Loughborough, UK) and water temperature (CS 547, Campbell Scientific) were measured continuously with a data logger from May 2003 to December 2003, and measurements were stored every 5 min (CR10X, Campbell Scientific). Exchange of DO with the atmosphere was calculated based on the O_2 saturation deficit or excess and on reaeration rates determined from the decline in dissolved propane concentrations during steady state injections of propane and Cl^- as a conservative tracer (Genereux and Hemond 1992, see detailed methods in Acuña et al. 2004).

Macroinvertebrate communities

Samples for macroinvertebrate community analysis were collected biweekly throughout the study and more frequently during periods of rapid change to characterize the boundaries between phases. Four replicate samples were collected from each available substratum (cobbles, sand, and leaf) on each of the 25 sampling dates. Cobbles were not sampled when they were out of the water. Medium cobbles (mean surface area = 778 cm^2) were removed from the streambed and washed immediately over a $250\text{-}\mu\text{m}$ sieve on a plastic tray. Invertebrates were collected from the sieve and fixed in 4% formaldehyde. Cobbles were wrapped with aluminum foil, and the foil was weighed to estimate cobble

surface area (regression determined empirically; $r^2 = 0.99$). Sand samples were collected with a 120-cm^2 corer to a depth of 3 cm. Leaf samples were collected with a 150-cm^2 corer.

Additional qualitative community samples were collected during the period of pool isolation to ensure that all taxa were detected. Community and physicochemical samples were collected from different pools during this period of pool isolation, but only data from the largest pool of the study reach are presented.

Macroinvertebrates were identified to the lowest possible taxonomic group and counted. Oligochaeta, Tipulidae, and Empididae were identified to family, whereas Nematoda and Hydracarina were not identified further. Samples containing ≥ 1000 individuals were subsampled by analyzing $\frac{1}{3}$ of the sample. Ash-free dry mass (AFDM) was determined for each identified taxon and for different size classes within each taxon (APHA 1981).

Statistical analysis

Mean density, biomass, taxon richness, and Shannon-Wiener diversity of all invertebrates were calculated for each sampling date and each substratum, and were weighted by substrate cover to express these metrics at a reach scale. Rare taxa (present in $< 5\%$ of the samples and making up $< 5\%$ of the total abundance) were omitted, leaving the common taxa for subse-

TABLE 1. Annual means of environmental variables measured from January 1997 to December 2003 at a meteorological station within the Fuirosos catchment.

	1997	1998	1999	2000	2001	2002	2003
Air temperature							
Mean monthly values (°C)	13.9	13.4	13.6	14.3	13.8	13.4	13.9
Mean monthly maxima (°C)	19.1	19.1	20.6	19.8	19.6	18.7	19.1
Mean monthly minima (°C)	10.5	9.5	11.3	10.8	10.3	10.1	10.5
Relative humidity							
Mean monthly values (%)	77.4	79.9	80.8	82.7	77.3	77.8	75.1
Mean monthly maxima (%)	93.4	95.8	96.6	95.9	92.7	93.3	90.6
Mean monthly minima (%)	57.2	56.9	59.0	62.8	56.2	58.2	55.6
Precipitation							
Total accumulated rainfall (L/m ²)	670	674	479	626	633	1071	814
Days with rain	121	90	36	110	99	137	114

quent analysis. Densities were $\log_{10}(x+1)$ transformed to reduce the effects of extreme values (Elliott 1977, Gauch 1982). The coefficient of variation (CV) was used to quantify physicochemical variability because CV adjusts the sample variance by the mean and, thus, is a better comparative measure of variability than sample variance alone (Palmer et al. 1997). Ordination was done using detrended correspondence analysis (DCA), a method considered appropriate in most community analysis applications (Wartenberg et al. 1987, Peet et al. 1988). DCAs were based on mean density on each sampling date and were done using CANOCO 4.0 (C. J. F. ter Braak and P. Smilauer. 1988. CANOCO reference manual and user's guide to Canoco for Windows: software for Canonical Community Ordination, version 4, Microcomputer Power, Ithaca, New York).

Similarity percentages (SIMPER) were used to determine the contribution of each species to the mean Bray–Curtis similarity within groups and to the dissimilarity between groups of community samples identified in the ordination. Analysis of similarities (ANOSIM), an analog of univariate analysis of variance (ANOVA), was used to test for differences between groups of community samples. SIMPER and ANOSIM

were done using PRIMER 5 (PRIMER E Ltd., Plymouth, UK).

Six ecological metrics were used in subsequent correlation analyses: mean density, mean richness, mean Shannon–Weiner diversity (H'), mean biomass, and the first 2 DCA axis scores (DCAAX1 and DCAAX2). Several environmental variables also were used in these correlation analyses. Mean daily water depth on the day of sampling and the change of depth in the 5 d before sampling were included to examine the short-term effect of drying and postdrought recovery. Water temperature and DO and NH_4 concentrations were used to characterize general physical and chemical conditions. Change in DO in the 5 d before sampling was used to examine the influence of O_2 depletion as a stressor and to evaluate the temporal component of O_2 depletion. Detritus and leaf coverage were used to represent organic matter storage dynamics, and cobble and sand coverage was used to represent the types of inorganic substrata present over time.

One-way ANOVAs were used to test differences between seasons and between dry (2001 and 2003) and wet years (2002). For most of these analyses, 2001 and 2003 data were combined and the combined 2001/2003 dry-year

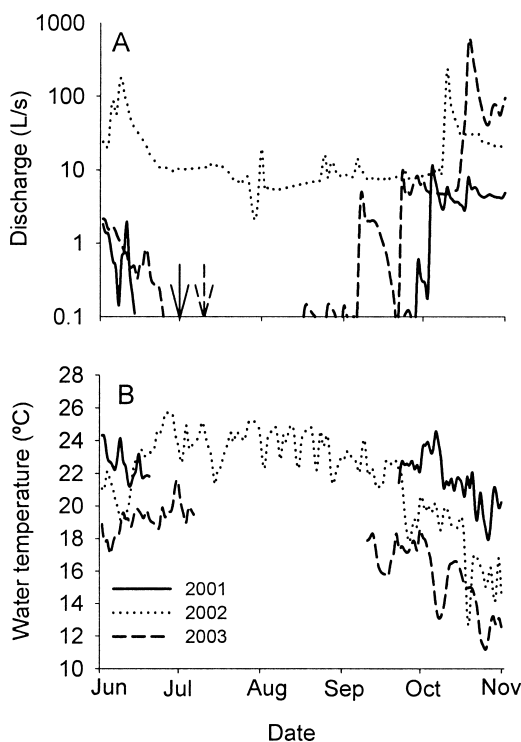


FIG. 3. Discharge (A) and water temperature (B) in the Fuirosos during the 3 study periods. Arrows indicate watercourse desiccation.

data were compared to the 2002 wet-year data. ANOVA and correlation analyses were done with STATISTICA (version 5.5, StatSoft, Tulsa, Oklahoma).

Results

Physicochemical changes

The Fuirosos dried from downstream to upstream. The headwater sections of the stream contained water during the entire study period, whereas the lower reaches dried during 2 of the 3 summers. Annual rainfall was low in 2001 (Table 1), and discharge declined steadily to the end of June when the stream dried (Fig. 3A). A few pools along the stream reach persisted for 2 wk until surface water disappeared completely. Flow resumed gradually in late September 2001. Annual rainfall was higher in 2002 than in 2001, and precipitation fell throughout summer. The stream continued to flow during summer 2002. Annual rainfall was relatively high in 2003 (Table 1), but precipitation during summer

was minimal and discharge declined until the end of June, with the same process of pool isolation described for 2001. Discharge resumed in a series of flow pulses that led to a swift flow recovery by mid September. The extent of intermittency (annual number of days having 0 flow) was 101 in 2001, 0 in 2002, and 68 in 2003.

Water temperature increased slightly during drought in 2003, whereas it persisted without abrupt changes throughout the dry season in 2002 (Fig. 3B). The end of summer was associated with a steady decrease in water temperatures in all years.

Watercourse fragmentation involved pool isolation and the onset of rapid physicochemical changes in 2001/2003. Declining water volume was associated with a steady increase in conductivity ($r = -0.92$, $p < 0.01$; Fig. 4) and a decrease in DO to hypoxic conditions (2.7 mg O_2 /L in 2001 and 0.44 mg O_2 /L in 2003) in pools. At the same time, soluble reactive P rose from 2.6 to 8.6 $\mu\text{g PO}_4\text{-P/L}$ and NH_4 increased from 36 to 120 $\mu\text{g NH}_4\text{-N/L}$.

The variability in several physicochemical features was higher in 2001/2003 than in 2002. For example, CVs of DO were 19% in 2002 and 51% in 2001/2003. However, interannual variability in water temperature was less marked than variability in DO (CV in 2001/2003 = 16% and CV in 2002 = 13%).

BOM

The amount of BOM on the wetted streambed during the study periods averaged 61 ± 13 g C/m^2 in 2001, 29 ± 41 g C/m^2 in 2002, and 58 ± 38 g C/m^2 in 2003. BOM increased in the pools when flow ceased and ranged from 252 to 408 g C/m^2 . BOM was significantly higher in 2001/2003 than in 2002 (1-way ANOVA, $F_{1,19} = 7.39$, $p < 0.05$; Fig. 5A–C)). BOM did not accumulate during the summer when flow was continuous in 2002 (Fig. 5B), but it did accumulate in 2001 and 2003 when the stream dried (Fig. 5A, C). Detrital coverage increased during drying and was strongly negatively correlated with DO ($r = -0.66$, $p < 0.001$) and depth ($r = -0.63$, $p < 0.001$). Leaf coverage was strongly correlated with NH_4 concentration ($r = 0.53$, $p < 0.01$) and water temperature ($r = -0.67$, $p < 0.001$).

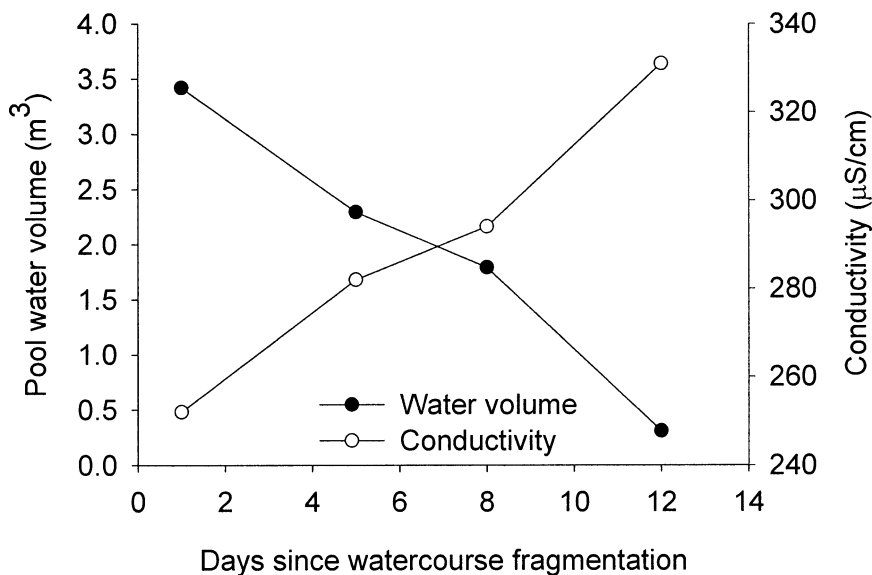


FIG. 4. Water volume and conductivity changes in the largest remaining pool after flow ceased in the Fuirosos in summer 2003.

Stream ecosystem metabolism

Daily metabolic rates differed in magnitude and pattern between 2002 and 2003 (Fig. 6A, B). During the postdrought recovery period, ER was higher in 2003 than in 2002 and followed a different temporal pattern. In 2002, ER and GPP increased during the summer and peaked in late September (Fig. 6A). In 2003, GPP decreased swiftly from ~ 0.8 to < 0.2 g O₂ m⁻² d⁻¹ during the drying period, whereas ER remained constant (Fig. 6B). Pool isolation led to a marked reduction in both GPP and ER, and postdrought recovery of ecosystem metabolism was stepped and followed discharge pulses. Ecosystem metabolism stabilized only with the onset of base-flow.

Invertebrate fauna

Seventy-six invertebrate taxa were found in the 25 surveys (Appendix). The number of taxa collected in any one survey ranged from 9 in July 2003 to 38 in October 2002. The number of species in the stream was undoubtedly > 76 because individuals in several taxonomic groups (e.g., Oligochaeta, Nematoda, and Hydracarina) were not identified to genus or species level. Insects made up 75% of the taxa collected, and 41% of the insects were Diptera. Chironomidae

was the most diverse family with 10 taxa. The most abundant taxon was Naididae (up to 1900 ind./m²). Mean density averaged 7927/m² in 2002 and 10,036 ind./m² in 2003, but this difference was not significant (1-way ANOVA, $F_{1,20} = 0.561$, $p > 0.05$; Fig. 7A). Neither richness (Fig. 7B) nor biomass (Fig. 7C) differed significantly between 2002 and 2003, but H' was significantly lower in 2003 than in 2002 (1-way ANOVA, $F_{1,20} = 10.25$, $p < 0.01$). H' averaged 2.33 in 2002 and 2.00 in 2003, and ranged from 1.25 at the end of June 2003 to 2.63 in October 2002.

Temporal patterns in density differed between 2002 and 2003 (Fig. 7A). In 2002, density peaked at 18,000 ind./m² at the beginning of July. In 2003, density peaked at 35,000 ind./m² following watercourse fragmentation at the end of June. In 2003, density decreased until the streambed was dry, whereas in 2002, density decreased and then increased to a 2nd peak in mid September. Overall, the seasonal signal of drying was stronger than interannual variability, i.e., density always increased during the contraction period, and density did not differ between years.

In general, the temporal patterns in biomass were similar to those of density ($r = 0.83$, $p < 0.01$; Fig. 7C). In 2003, mean biomass was significantly greater before cessation of flow than

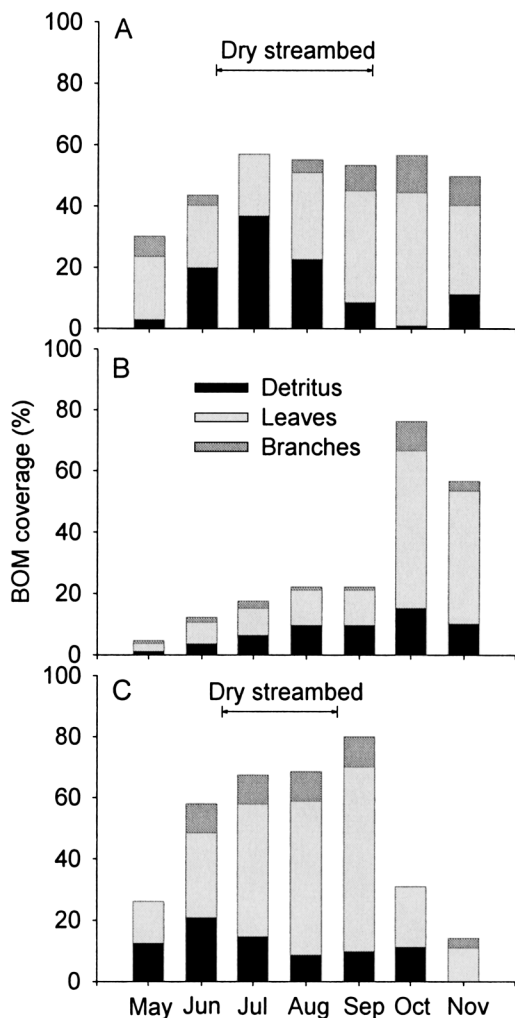


FIG. 5. Benthic organic matter (BOM) cover in the Fuirosos during 2001 (A), 2002 (B), and 2003 (C).

after flow recovery (1-way ANOVA, $F_{1,7} = 5.75$, $p < 0.05$), but richness and H' did not differ between phases. Density and biomass were negatively correlated with depth and positively correlated with detritus cover ($r = 0.88$, $p < 0.001$).

H' was significantly correlated with change of depth in the 5 d before sampling, i.e., with flow reduction during drying ($r = -0.36$, $p < 0.05$). H' was negatively correlated with PO_4 ($r = -0.21$, $p < 0.05$) and NH_4 concentrations ($r = -0.31$, $p < 0.05$). Maximum H' was observed at intermediate water levels (Fig. 8). The average Bray–Curtis similarity of the community samples was 26% in 2001/2003 and 42% in 2002.

Thus, heterogeneity in biotic communities was more pronounced in drier years.

Ordination distinguished broadly between communities from 2001/2003 and 2002 (Fig. 9). Axis 1 scores were significantly correlated with change of depth in the 5 d before sampling, PO_4 concentrations, NH_4 concentrations, leaf cover, and mean diversity. Axis 2 scores were significantly correlated with water temperature ($r = -0.71$) and depth ($r = 0.47$). Changes along the 1st axis appeared to be the result of drying, whereas changes along the 2nd axis appeared to be the result of postdrought recovery.

The change in community structure during drying could be seen as a shift in scores on the first 2 axes along a diagonal from the upper left to the lower right on the ordination plot in both 2001/2003 and 2002 (Fig. 9). As baseflow resumed, the community trajectory changed to form an anticlockwise hysteresis. The main differences between trajectories in 2001/2003 and 2002 were the amplitude of the variation in macroinvertebrate community structure and the differences during the recovery phase. In 2001/2003, the onset of flow was associated with significant changes in community structure. Predrought community samples were dominated by Oligochaeta (Propaidae and Naididae), whereas postdrought samples were dominated by Diptera (Chironomidae and Culicidae). In contrast, differences between pre- and postdrought community samples were minimal in 2002 (Fig. 9).

Macroinvertebrate community structure in relation to substrata

Macroinvertebrate densities were >2 orders of magnitude higher on sand and leaf substrata than on cobbles (1-way ANOVA, $F_{1,62} = 24.12$, $p < 0.0001$; Table 2). No consistent interannual differences in invertebrate density were found on sand or cobble substrata. However, mean density on leaves was significantly lower (8949 ind./m²) in 2001/2003 than in 2002 (21,805 ind./m²) (1-way ANOVA, $F_{1,16} = 4.98$, $p < 0.05$). Average Bray–Curtis similarities were 30% in cobbles, 29% in sand, and 22% in leaves. Average Bray–Curtis dissimilarities were 96% between cobbles and sand, 92% between cobbles and leaves, and 82% between sand and leaves (Table 2). Macroinvertebrate community composition differed considerably within each substratum but the differences among substrata

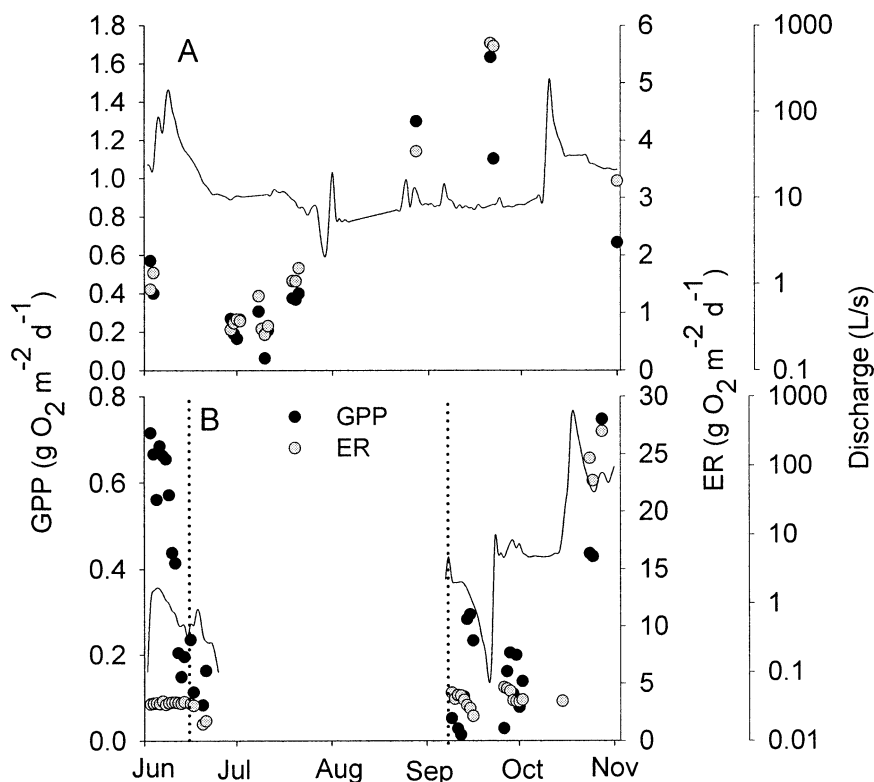


FIG. 6. Discharge (solid lines) and daily rates of whole-stream metabolism gross primary production (GPP) and ecosystem respiration (ER) in the Fuirosos during the study periods of 2002 (A) and 2003 (B). Vertical dotted lines in 2003 indicate dates when flow ceased and resumed.

were higher than differences within substrata (Table 2).

Discussion

Effect of drought

The telescoping ecosystem model of Fisher et al. (1998), modified by Ward and Tockner (2001), predicts maximum heterogeneity in habitat features and biotic communities during the dry phase. Temporal variation in habitat and biotic communities in the Fuirosos followed the patterns described by the telescoping ecosystem model. Flow reduction decreased the capacity of the stream to transport organic matter and caused an increase in detritus coverage and an increase in pool extent, thereby enhancing spatial habitat heterogeneity. However, flow cessation led to watercourse fragmentation into a series of isolated pools and caused an abrupt change in physicochemical conditions that im-

posed a threshold on the ecosystem. Organic matter and sediments accumulated in pools when flow ceased and organic matter transport stopped. The huge increase in the amount of detritus and reduced physical reaeration caused a decrease in DO concentrations and an increase in nutrient concentrations, typical responses to pool isolation (Townsend 1985, Stanley et al. 1997, Caruso 2002). As a consequence, variation in most physicochemical features was greater during the 2 dry years than during the wet year, a pattern similar that described by Williams and Hynes (1976).

Flow reduction also changed stream metabolism. GPP decreased during the drying phase, but ER did not change until the stream underwent watercourse fragmentation, when ER decreased sharply. Changes in the availability of detritus during drying may have controlled the macroinvertebrate community. Increasing availability of food (detritus) for invertebrates during

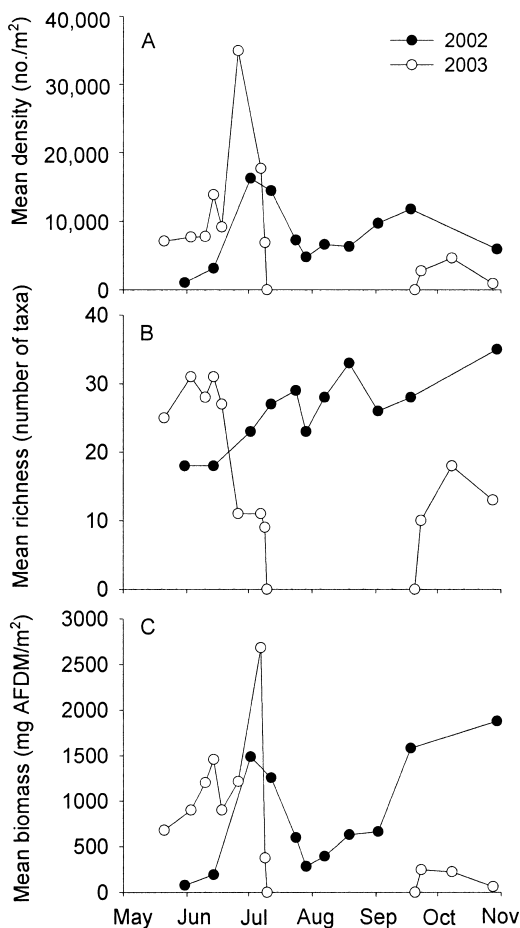


FIG. 7. Mean invertebrate density (A), richness (B), and biomass (C) in the Fuirosos in 2002 and 2003. AFDM = ash-free dry mass.

flow reduction can lead to a considerable increase in invertebrate density and biomass. In 2001/2003, Naididae, Lumbriculidae, Ostracoda, and *Chironomus* sp. were the main sources of the increase in invertebrate biomass and density, whereas in 2002, *Baetis* sp., *Corynoneura* sp., *Thienemanniella* sp., and *Simulium* sp. were the main sources of increased invertebrate density. Thus, the nature of the increase in invertebrate density and biomass during the drying process differed between wet and dry summers.

Low flows were accompanied by low DO concentrations. Density had increased by the end of the drying phase, but H' had decreased, suggesting that the physicochemical conditions in the stream became limiting for some species. Taxa with low DO requirements (e.g., Naididae,

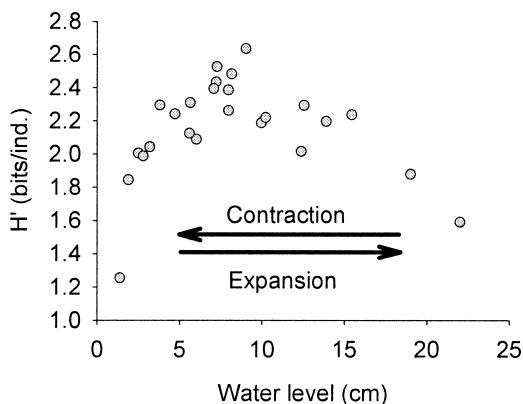


FIG. 8. Macroinvertebrate Shannon-Wiener diversity (H') vs water level in the Fuirosos during the study period.

Lumbriculidae, *Chironomus* sp., and adult Dytiscidae and Hydraenidae beetles) increased significantly as flow decreased, but taxa with high DO requirements (e.g., *Ancyclus fluviatilis* and *Bythinia* sp.) were absent during this phase. Bunn et al. (1986) reported that warm water and low flows may have been limiting for many species in Australian Mediterranean streams, but that these conditions favored chironomids and gave rise overall to increased densities and low diversity. The rapid increase of these tolerant species led to a density peak soon after flow cessation in the Fuirosos, a pattern also described by Towns (1985) and Boulton and Lake (1992b), but this density peak dropped rapidly in response to changes caused by flow cessation. In contrast, Smith and Pearson (1987) attributed the density peak that occurred soon after flow cessation in an intermittent stream in Tropical Queensland to congregation of mobile organisms in an ever-diminishing habitat. Regardless of the cause of peak densities during drying, physicochemical conditions shift rapidly with flow cessation and undoubtedly have adverse effects on the benthos. The most probable causes of these adverse effects are deoxygenation (Stanley et al. 1997, Tramer 1977, Labbe and Fausch 2000, Mol et al. 2000) and toxicity of certain leachates from leaf decomposition (Towns 1985, 1991, Boulton and Lake 1990, 1992b, Cherqui et al. 1997).

Another consequence of pool isolation in the Fuirosos was an increase in predation pressure (e.g., *Stictotarsus* sp., *Notonecta maculata*, *Cordu-*

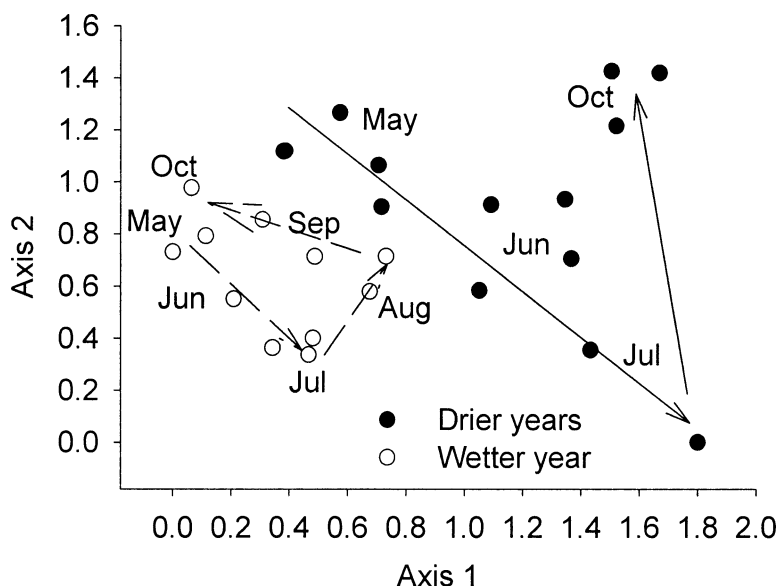


FIG. 9. Detrended correspondence analysis (DCA) of community samples by based on mean total densities of macroinvertebrates from the Fuirosos. Arrows indicate temporal dynamics.

TABLE 2. Invertebrate densities on different substrata, total Bray–Curtis dissimilarities between substrata, and % contribution of specific taxonomic groups to the Bray–Curtis dissimilarities between invertebrate communities in different substrata.

Taxon	Mean density (ind./m ²)		Mean Bray– Curtis dis- similarity (%)
	Cobbles	Sand	
Total			96
Tubificidae	1.3	3353	16.7
Naididae	2.4	3390	13.5
<i>Micropsectra</i> sp.	28.8	2486	10.7
Propaidae	0.7	2031	7.6
Total			92
	Cobbles	Leaves	
<i>Polypedilum</i> sp.	15.6	1859	13.2
<i>Rheocricotopus</i> sp.	50	2455	10.7
Naididae	2.4	796	8.6
<i>Ancyclus fluviatilis</i>	446	840	8.4
Total			82
	Sand	Leaves	
Tubificidae	3353	345	10.3
Naididae	3390	796	10.2
<i>Micropsectra</i> sp.	2486	845	7.9
<i>Polypedilum</i> sp.	405	1859	6

legaster sp.). Densities of predators rise sharply during drying in many temporary streams (Stehr and Branson 1938, Hynes 1975, Abell 1984, Stanley et al. 1994). Fragmentation of the Fuirosos into isolated pools was accompanied by a decrease in mean density, the replacement of many rheophilous taxa by taxa more commonly found in lentic habitats (e.g., veliids and dytiscids; Boulton and Lake 1992b), and an increase in predation pressure (Boulton and Lake 1992b, Stanley et al. 1994, Closs and Lake 1996).

Postdrought recovery

The extent of intermittency strongly controls organic matter storage because organic matter accumulates on the dry streambed (Boulton and Lake 1992b, Acuña et al. 2004). In the Fuirosos, large amounts of organic matter were available at the end of the dry seasons in 2001 and 2003, but continuous flow during 2002 minimized leaf accumulation. Therefore, organic matter storage in summer differed between years depending on whether flow ceased (Acuña et al. 2004), and differences in the amount of BOM were correlated with differences in physicochemical conditions and ER between 2001/2003 and 2002.

Flow maintained macroinvertebrate community structure throughout the summer of 2002,

and pre- and postdrought communities were similar. The similarity was reflected by the continuous presence of species associated with high water velocities (e.g., *Baetis* sp. and *Simulium* sp.). However, in 2001/2003, community structure changed substantially from the predrought to the postdrought communities. Flow recovery was steady in 2001, whereas recovery was stepped and sharp in 2003. Unfortunately, data from 2001 are scarce, and a detailed analysis of the postdrought recovery period is possible only for autumn 2003. In autumn 2003, the community consisted mostly of pioneer species such as dipteran larvae (mainly Chironomidae and Ceratopogonidae), and densities were an order of magnitude lower than in spring 2003. This pioneer community structure is typical of macroinvertebrate communities after rewetting (Extence 1981, Stanley et al. 1994). The difference in community structure between autumns of wet and dry years strongly suggests that watercourse desiccation may influence community development during the year following the drought, a long-term impact of drought that has been reported by Boulton and Lake (1992a).

The threshold effect of watercourse desiccation on community composition in the Fuirosos implies that few taxa showed resistance during drying. As in other intermittent streams (Boulton and Lake 1992b, Shivoga 2001), most taxa in the Fuirosos had no specialized adaptations for surviving intermittence. Thus, resilience appears to be the predominant strategy for surviving drought in the Fuirosos. Other authors have stressed the importance of resilience rather than resistance in desert streams (Harrison 1966, Hynes 1975, Gagneur and Chaoui-Boudghane 1991, Stanley et al. 1994), and our work extends these results to Mediterranean streams.

Overall, maximum heterogeneity in habitat features and biotic communities was associated with low flows during the drying phase in the Fuirosos, but the influence of flow dynamics on habitat heterogeneity varies among geomorphic stream types (Ward and Tockner 2001). Our knowledge about relationships between habitat heterogeneity and water level is derived mainly from research carried out in European alluvial floodplains (Schiemer et al. 1999, Malard et al. 2000, Ward and Tockner 2001) or in tropical rivers (Mertes et al. 1995). The constrained character of the channel and the streambed mor-

phology in the Fuirosos may have been responsible for the increased habitat heterogeneity and resulting diversity at low flows. However, the modified telescoping ecosystem model (Fisher et al. 1998, Ward and Tockner 2001) seems applicable for explaining variation of structural and functional ecosystem components during the drought and postdrought recovery phases in this Mediterranean stream.

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APPENDIX. Macroinvertebrate taxa from the Fuirosos.

Taxonomic group	Genus/species
Oligochaeta	
Enchytraeidae	
Lumbricidae	
Lumbriculidae	
Propaidae	
Tubificidae	
Naididae	
Hirundinia	
Erpobdellidae	<i>Erpobdella</i> sp.
Glossiphoniidae	<i>Glossiphonia</i> sp.
Mollusca	
Ancylidae	<i>Ancylus fluviatilis</i>
Lymnaeidae	<i>Radix peregra</i>
	<i>Galba truncatula</i>
Physidae	<i>Physa</i> sp.
Bythinellidae	<i>Bythinia</i> sp.
Hydracarina	
Ephemeroptera	
Baetidae	<i>Baetis</i> sp.
	<i>Cloeon dipterum</i>
Ephemerellidae	<i>Ephemerella ignita</i>
Leptophlebiidae	<i>Habrophlebia fusca</i>
	<i>Paraleptophlebia cincta</i>
Heptageniidae	<i>Ecdyonurus tenosus</i>
Caenidae	<i>Caenis luctosa</i>
Plecoptera	
Capniidae	<i>Capnia bifrons</i>
	<i>Capnioneura mitis</i>
Nemouridae	<i>Nemoura lacustris</i>
Perlodidae	<i>Isoperla grammatica</i>
Leuctridae	
Trichoptera	
Hydropsychidae	<i>Hydropsyche siltalai</i>
Limnephilidae	<i>Glyptotaelius pellucidus</i>
	<i>Limnephilus guadarramicus</i>
Rhyacophilidae	<i>Rhyacophila</i> sp.
Leptoceridae	<i>Mystacides azurea</i>
Polycentropodidae	<i>Polycentropus kingi</i>
	<i>Plectrocnemia</i> sp.
Diptera	
Ceratopogonidae	Ceratopogoninae
Chironomidae	<i>Chironomus</i> sp.
	<i>Corynoneura</i> sp.
	<i>Thienemannimyia</i> sp.
	<i>Micropsectra</i> sp.
	<i>Orthocladius</i> sp.
	<i>Polypedilum</i> sp.
	<i>Prodiamesa</i> sp.
	<i>Rheocricotopus</i> sp.
	<i>Thienemanniella</i> sp.
	<i>Tvetenia</i> sp.

APPENDIX. Continued.

Taxonomic group	Genus/species
Culicidae	
Dixidae	<i>Dixa</i> sp.
Empididae	
Limoniidae	<i>Dicranota</i> sp.
	<i>Hexatoma</i> sp.
Rhagionidae	<i>Chrysopilus</i> sp.
Scatophagidae	<i>Acanthocnema</i> sp.
Sciomyzidae	
Simuliidae	<i>Simulium</i> sp.
Tabanidae	
Tipulidae	
Coleoptera	
Dryopidae	<i>Dryops</i> sp.
Dytiscidae	<i>Hydroporus</i> sp.
	<i>Deronectes</i> sp.
	<i>Stictotarsus</i> sp.
Elmidae	<i>Oulimnius</i> sp.
Hydrophilidae	<i>Anacaena</i> sp.
Scirtidae	<i>Cyphon</i> sp.
Odonata	
Anisoptera	<i>Onychogomphus</i> sp.
Zygoptera	<i>Calopteryx</i> sp.
Heteroptera	
Gerridae	<i>Aquaris najas</i>
	<i>Gerris lacustris</i>
Hydrometridae	<i>Hydrometra stagnorum</i>
Mesoveliidae	<i>Mesovelia</i> sp.
Nepidae	<i>Nepa cinerea</i>
Notonectidae	<i>Notonecta pallidula</i>
	<i>Notonecta maculata</i>
Veliidae	<i>Velia caprai</i>
	<i>Microvelia reticulata</i>
Crustacea	
Branchiopoda	<i>Alona</i> sp.
Ostracoda	
Nematoda	